

Signorini's Problem

Max Stuthmann

March 13, 2026

1 Introduction

2 Sobolev Spaces

- Integer Order Sobolev Spaces
- Fractional Order Sobolev Spaces
- Trace Operators and Boundary Spaces
- $H(\operatorname{div}; \Omega)$ and Normal Traces

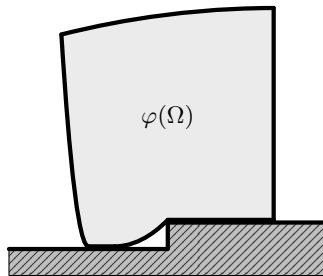
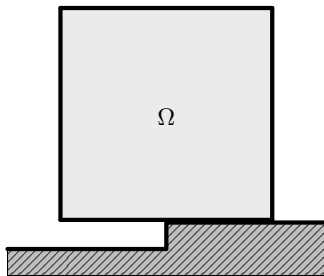
3 Signorini's Problem

- Geometric Setup
- Weak Formulations
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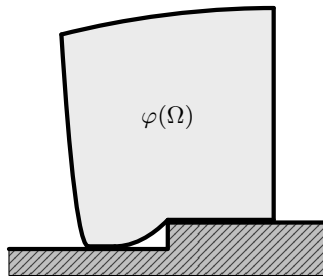
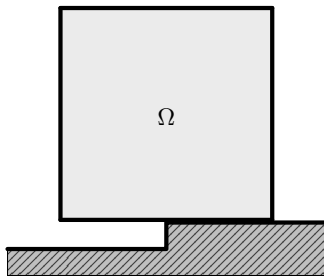
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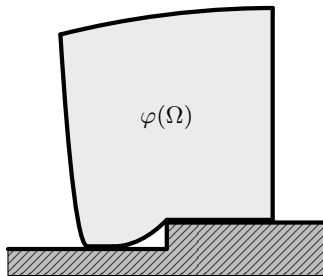
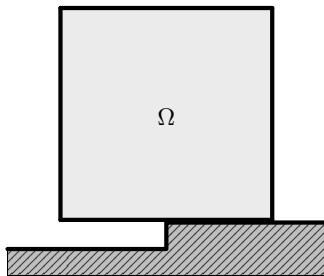
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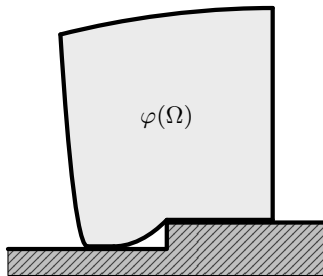
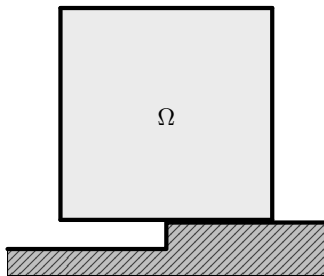
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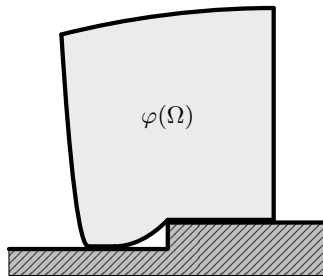
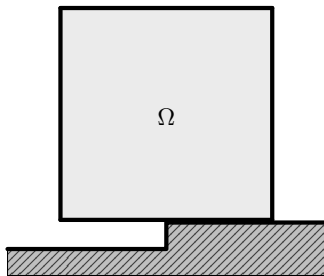
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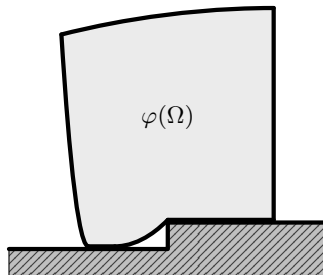
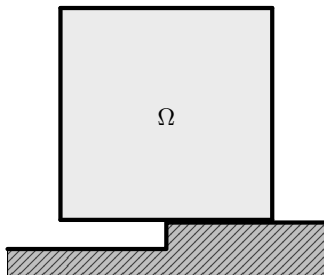
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- ▶ Complicated part: Contact Boundary.
- ▶ Rigid foundation can be described by gap-function g but we assume a flat foundation, i.e. $g = 0$.



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- ▶ For $k = 1$ we get

$$\|u\|_{1,\Omega} = \left(\|u\|_{0,\Omega}^2 + \|\nabla u\|_{0,\Omega}^2 \right)^{1/2}, \quad u \in H^1(\Omega; \mathbb{R}^m).$$

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- ▶ Boundary traces of H^k -functions live in H^s spaces.

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$$H^{s-1/2}(\Gamma; \mathbb{R}^m) := \gamma_0(\mathcal{C}^\infty(\overline{\Omega}; \mathbb{R}^m))$$

and equip it with the norm

$$\|g\|_{s-1/2, \Gamma} := \inf \left\{ \|u\|_{s, \Omega} \mid u \in \mathcal{C}^\infty(\overline{\Omega}; \mathbb{R}^m), \gamma_0 u = g \right\}.$$

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- ▶ Same set for all $s > 1/2$ but different norms.

The Trace Theorem

Theorem (Trace Theorem)

Let $1/2 < s \leq 1$. Then the classical trace operator γ_0 extends uniquely to a continuous surjective linear operator

$$\gamma : H^s(\Omega; \mathbb{R}^m) \rightarrow H^{s-1/2}(\Gamma; \mathbb{R}^m).$$

In particular, there exists a constant $C_{\text{tr}} > 0$ such that

$$\|\gamma u\|_{s-\frac{1}{2}, \Gamma} \leq C_{\text{tr}} \|u\|_{s, \Omega} \quad \forall u \in H^s(\Omega; \mathbb{R}^m).$$

In the special case $s = 1$, one has

$$\ker(\gamma) = \overline{\mathcal{D}(\Omega; \mathbb{R}^m)}^{\|\cdot\|_{1, \Omega}} =: H_0^1(\Omega; \mathbb{R}^m).$$

Boundary Spaces on Boundary Parts

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with norm

$$\|g\|_{s-\frac{1}{2}, \Gamma_0} := \inf \left\{ \|\tilde{g}\|_{s-\frac{1}{2}, \Gamma} \mid \tilde{g} \in H^{s-1/2}(\Gamma; \mathbb{R}^m), \tilde{g}|_{\Gamma_0} = g \right\}.$$

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- ▶ Moreover, we define

$$H_{00}^{s-1/2}(\Gamma_0; \mathbb{R}^m) := \left\{ g \in H^{s-1/2}(\Gamma_0; \mathbb{R}^m) \mid \text{Ext}_0(g) \in H^{s-1/2}(\Gamma; \mathbb{R}^m) \right\},$$

where $\text{Ext}_0(g)$ denotes the zero extension of g from Γ_0 to Γ with the norm

$$\|g\|_{s-\frac{1}{2}, 00, \Gamma_0} := \|\text{Ext}_0(g)\|_{s-\frac{1}{2}, \Gamma}.$$

Dual Spaces of Boundary Spaces

► We define

$$H^{-1/2}(\Gamma; \mathbb{R}^m) := (H^{1/2}(\Gamma; \mathbb{R}^m))', \quad H^{-1/2}(\Gamma_0; \mathbb{R}^m) := (H_{00}^{1/2}(\Gamma_0; \mathbb{R}^m))'.$$

with duality pairings $\langle \cdot, \cdot \rangle_\Gamma$ and $\langle \cdot, \cdot \rangle_{\Gamma_0}$, respectively.

Theorem (Lifting Operator)

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $\Gamma_0 \subset \Gamma$ be relatively open.

- i) There exists a bounded linear operator $\mathcal{L} : H^{1/2}(\Gamma; \mathbb{R}^d) \rightarrow H^1(\Omega; \mathbb{R}^d)$ such that

$$\gamma(\mathcal{L}g) = g \quad \forall g \in H^{1/2}(\Gamma; \mathbb{R}^d).$$

- ii) There exists a bounded linear operator $\mathcal{L} : H_{00}^{1/2}(\Gamma_0; \mathbb{R}^d) \rightarrow H^1(\Omega; \mathbb{R}^d)$ such that for every $g \in H_{00}^{1/2}(\Gamma_0; \mathbb{R}^d)$,

$$\gamma(\mathcal{L}g)|_{\Gamma_0} = g \quad \text{and} \quad \gamma(\mathcal{L}g) = 0 \text{ on } \Gamma \setminus \Gamma_0.$$

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$H(\operatorname{div}; \Omega)$ for Matrix-Valued Fields

- ▶ The space $\operatorname{Sym}^2(\mathbb{R}^d)$ of symmetric second order tensors is

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- ▶ On $\mathbb{R}^{d \times d}$, we use the Frobenius product

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- ▶ If $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\operatorname{Sym}^2(\mathbb{R}^d); \operatorname{Sym}^2(\mathbb{R}^d))$ is a fourth order tensor and $s \in \operatorname{Sym}^2(\mathbb{R}^d)$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{kl}, \quad s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- ▶ We define

$$H(\operatorname{div}; \Omega) := \{\tau \in L^2(\Omega; \operatorname{Sym}^2(\mathbb{R}^d)) \mid \operatorname{div} \tau \in L^2(\Omega; \mathbb{R}^d)\},$$

where $\operatorname{div} \tau$ is understood row-wise in the weak sense.

Normal Traces and Boundary Traction

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$$\gamma v \in H^{1/2}(\Gamma; \mathbb{R}^d) \subset L^2(\Gamma; \mathbb{R}^d).$$

Hence the normal component

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- ▶ This leads naturally to $\tau \in H(\operatorname{div}; \Omega)$.
- ▶ On $H(\operatorname{div}; \Omega)$, the trace theorem does not hold. We need a different concept here.

Normal Traces and Boundary Traction

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- ▶ In the smooth case, $\gamma_n(\tau)$ coincides with the classical traction $\tau \mathbf{n}$. We thus write $\tau \mathbf{n} = \gamma_n(\tau)$.

Normal Traces on Boundary Parts

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- ▶ Let $\Gamma_0 \subset \Gamma$ be open in Γ .
- ▶ The normal trace on Γ_0 is defined as the unique element $\gamma_{n,0}(\tau) \in H^{-1/2}(\Gamma_0; \mathbb{R}^d)$ given by

$$\langle \gamma_{n,0}(\tau), g \rangle_{\Gamma_0} := \langle \gamma_n(\tau), \text{Ext}_0(g) \rangle_{\Gamma}, \quad g \in H_{00}^{1/2}(\Gamma_0; \mathbb{R}^d),$$

where $\text{Ext}_0(g)$ denotes the zero extension of g from Γ_0 to Γ .

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Boundary Decomposition

- ▶ Elastic Body $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$ open and bounded domain with Lipschitz boundary $\Gamma := \partial\Omega$.
- ▶ Decompose Γ into three relatively open and pairwise disjoint parts

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Boundary Decomposition

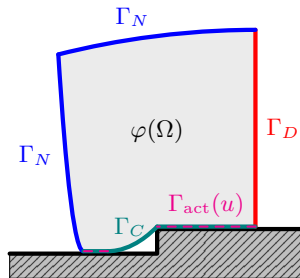
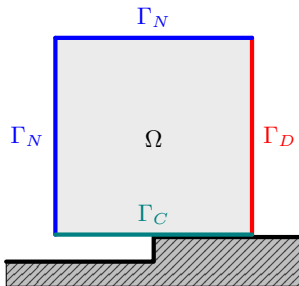
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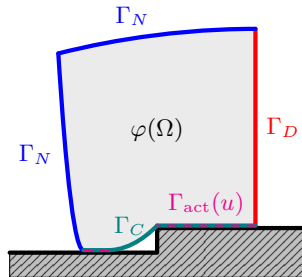
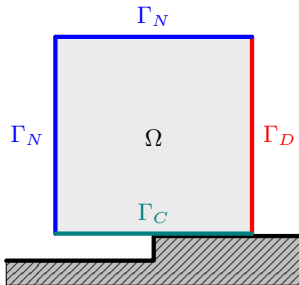
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- Each part has different boundary conditions:



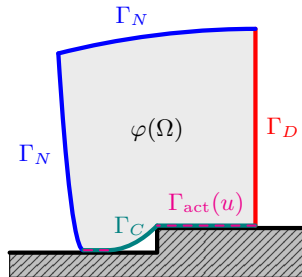
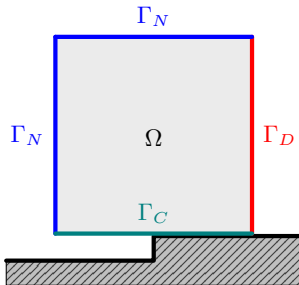
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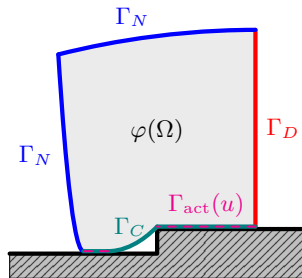
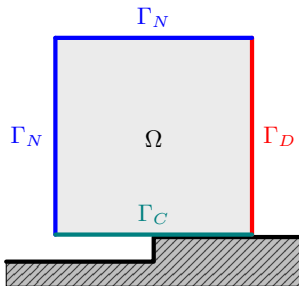
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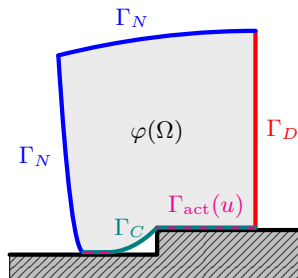
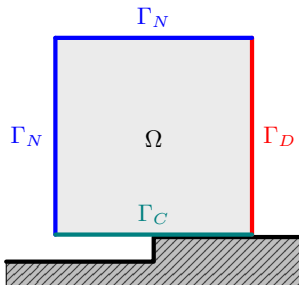
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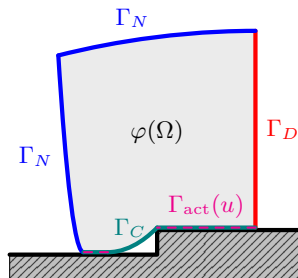
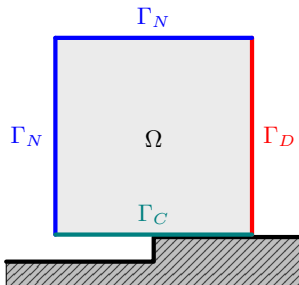
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- *Small strain tensor:*

$$\varepsilon : H^1(\Omega; \mathbb{R}^d) \rightarrow L^2(\Omega; \text{Sym}^2(\mathbb{R}^d)), \quad u \mapsto \varepsilon(u) := \frac{1}{2} (\nabla u + \nabla u^\top).$$

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$$s : \mathcal{A}(x) : s \geq \alpha_{\mathcal{A}} (s : s) \quad \forall s \in \text{Sym}^2(\mathbb{R}^d).$$

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$$|\mathcal{A}_{ijkl}(x)| \leq C_{\mathcal{A}} \quad \forall 1 \leq i, j, k, l \leq d.$$

- ▶ *(Cauchy) stress tensor:*

$$\sigma : H^1(\Omega; \mathbb{R}^d) \rightarrow L^2(\Omega; \text{Sym}^2(\mathbb{R}^d)), \quad u \mapsto \mathcal{A} : \varepsilon(u).$$

Virtual Work: External and Internal Forces

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Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : V \times V \rightarrow \mathbb{R}$ defined above is

i) bounded, i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \quad \text{with} \quad C_1 = C_A,$$

ii) V -elliptic, i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2 \quad \text{with} \quad C_2 = \frac{\alpha_A}{C_K^2}.$$

Moreover, the linear functional ℓ defined above is bounded, i.e. for all $v \in V$ there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \quad \text{with} \quad C_3 = \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N})$$

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Lemma (Weak lower semicontinuity of the norm)

Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \rightarrow \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X , then

$$\|x\|_X \leq \liminf_{j \rightarrow \infty} \|x_j\|_X.$$

Proof: Omitted here.

Existence of Minimizer

Lemma ($\mathcal{J}$ is coercive and w.l.s.c.)

The functional $\mathcal{J}$ is

i) coercive on V , i.e.

$$\mathcal{J}(v) \rightarrow \infty \quad \text{as} \quad \|v\|_{1,\Omega} \rightarrow \infty,$$

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► By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'} \|v\|_{1,\Omega}$. Hence

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► This proves i).

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- ▶ Therefore,

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \leq \frac{1}{2} \liminf_{j \rightarrow \infty} a(v_j, v_j) - \lim_{j \rightarrow \infty} \ell(v_j) = \liminf_{j \rightarrow \infty} \mathcal{J}(v_j).$$



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The functional $\mathcal{J}$ has at least one minimizer $u \in K$.

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- ▶ By w.l.s.c. of $\mathcal{J}$, $u \in K$ is minimizer of $\mathcal{J}$ because

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Proposition (Uniqueness of a Minimizer of $\mathcal{J}$)

If there exists a minimizer of the functional $\mathcal{J}$, it is unique.

Proof: $\mathcal{J}$ is strictly convex on V and thus on the convex set K . Thus, a minimizer $u \in K$ is unique. $\square$

A-Priori Stability Bound

Proposition (Stability estimate for the minimizer)

The Unique minimizer $u \in K$ of $\mathcal{J}$ satisfies the a-priori bound

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- ▶ Finally,

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Variational Inequality as an Optimality Condition

- ▶ Fréchet Derivative of $\mathcal{J}$ at $u \in V$ in direction $h \in V$ is $\mathcal{J}'(u)[h] = a(u, h) - \ell(h)$.

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A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

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- ▶ By chainrule, $\phi'(0^+) = \mathcal{J}'(u)[v - u] \geq 0$.

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- ▶ $u \in K$ minimizes $\mathcal{J}$ over K . □

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Another Stability Estimate

Proposition (Lipschitz Dependence on the Data)

For $i \in \{1, 2\}$, let $f_i \in L^2(\Omega; \mathbb{R}^d)$ and $F_i \in L^2(\Gamma_N; \mathbb{R}^d)$, and let $\ell_i \in V'$ be the corresponding load functional, i.e.

$$\ell_i(v) := \int_{\Omega} f_i \cdot v \, dx + \int_{\Gamma_N} F_i \cdot \gamma v \, ds, \quad v \in V.$$

Moreover, let $u_i \in K$ be the unique solution of one of the weak formulations above. Then the solution depends Lipschitz continuously on the load functional, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2}{\alpha_{\mathcal{A}}} \|\ell_1 - \ell_2\|_{V'}.$$

In particular, it depends Lipschitz continuously on the data f_i, F_i in the sense that

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2 \max\{1, C_{\text{tr}}\}}{\alpha_{\mathcal{A}}} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \right).$$

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2 Sobolev Spaces

3 Signorini's Problem

- Geometric Setup
- Weak Formulations
- **Strong Formulations**
- Mixed Formulations

4 Further Topics

Stress Regularity

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$$\varepsilon(u) \in L^2(\Omega; \operatorname{Sym}^2(\mathbb{R}^d)) \quad \text{and} \quad \sigma(u) = \mathcal{A} : \varepsilon(u) \in L^2(\Omega; \operatorname{Sym}^2(\mathbb{R}^d)).$$

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- ▶ Therefore Green's formula is

$$\langle \sigma(u)n, \gamma v \rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v) \, dx + \int_\Omega \operatorname{div} \sigma(u) \cdot v \, dx \quad \forall v \in H^1(\Omega; \mathbb{R}^d).$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$, find $u \in K$ such that $\sigma(u) \in H(\text{div}; \Omega)$ and

$$\left\{ \begin{array}{ll} -\text{div } \sigma(u) = f & \text{a.e. in } \Omega, \\ \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot \gamma u \, ds, \\ \langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma v \, ds \quad \forall v \in K. \end{array} \right. \quad \begin{array}{l} (1a) \\ (1b) \\ (1c) \end{array}$$

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$. Then $u \in K$ solves the hybrid problem above if and only if it solves the weak problem

$$\text{Find } u \in K \text{ such that} \quad a(u, v - u) \geq \ell(v - u) \quad \forall v \in K.$$

Key Identity for Proof

- ▶ Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$ and $-\operatorname{div} \sigma(u) = f$ in $L^2(\Omega; \mathbb{R}^d)$.

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$$\langle \sigma(u)n, \gamma w \rangle_\Gamma = a(u, w) + \int_\Omega \operatorname{div} \sigma(u) \cdot w \, dx = a(u, w) - \int_\Omega f \cdot w \, dx.$$

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- ▶ Hence for all $w \in V$:

$$a(u, w) - \ell(w) = \langle \sigma(u)n, \gamma w \rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma w \, ds.$$

Proof: Hybrid $\Rightarrow$ Weak

- ▶ Subtract (1b) from (1c):

$$\langle \sigma(u)\mathbf{n}, \gamma(v - u) \rangle_{\Gamma} \geq \int_{\Gamma_N} F \cdot \gamma(v - u) \, ds.$$

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- ▶ Insert the key identity from the previous slide:

$$a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K,$$

i.e. the weak formulation.

Proof: Weak $\Rightarrow$ Hybrid (Bulk Equation)

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- ▶ Thus, $-\operatorname{div} \sigma(u) = f$ in $\mathcal{D}'(\Omega; \mathbb{R}^d)$ and by $\sigma(u) \in H(\operatorname{div}; \Omega)$ we have

$$-\operatorname{div} \sigma(u) = f \quad \text{a.e. in } \Omega.$$

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- ▶ With the key identity this is exactly (1b) since

$$0 = a(u, u) - \ell(u) = \langle \sigma(u)\mathbf{n}, \gamma u \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma u \, ds.$$

Proof: Weak $\Rightarrow$ Hybrid (Hybrid Equality)

► Let $v \in K$. We have

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- ▶ This means

$$\langle \sigma(u)n, \gamma v \rangle_{\Gamma} \geq \int_{\Gamma_N} F \cdot \gamma v \, ds + \underbrace{\left(\langle \sigma(u)n, \gamma u \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma u \, ds \right)}_{=0}.$$



Normal/Tangential Splitting and Contact Trace Spaces

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- ▶ Every $w \in W_C$ admits the decomposition

$$w = w_n \mathbf{n} + w_t \in W_{C,n} \mathbf{n} \oplus W_{C,t}$$

Normal/Tangential Splitting and Contact Trace Spaces

- ▶ Dual Spaces W'_C , $W'_{C,n}$, $W'_{C,t}$ and duality pairing $\langle \cdot, \cdot \rangle_{\Gamma_C}$.

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of admissible normal traces.

- ▶ Dual cone of weakly nonpositive normal contact forces:

$$\Lambda_C := \left\{ \lambda \in W'_{C,n} \mid \langle \lambda, \eta \rangle_{\Gamma_C} \geq 0 \quad \forall \eta \in W_{C,n}^- \right\}.$$

Karush-Kuhn-Tucker Contact Conditions

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For given $f \in L^2(\Omega; \mathbb{R}^d)$, find $u \in V$ such that

$$\left\{ \begin{array}{ll} -\operatorname{div} \sigma(u) = f & \text{a.e. in } \Omega, \\ \sigma_t(u) = 0 & \text{in } W'_{C,t}, \\ u_n \leq 0 & \text{a.e. on } \Gamma_C, \\ \sigma_n(u) \in \Lambda_C, \\ \langle \sigma_n(u), u_n \rangle_{\Gamma_C} = 0. \end{array} \right. \quad \begin{array}{l} (2a) \\ (2b) \\ (2c) \\ (2d) \\ (2e) \end{array}$$

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- ▶ Assuming $|\Gamma_N| = 0$, this is equivalent to previous formulations.

Classical Strong Formulation under Additional Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ find $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations

$$\left\{ \begin{array}{ll} -\operatorname{div} \sigma(u) = f & \text{in } \Omega, \\ \sigma(u) = \mathcal{A} : \varepsilon(u) & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma_D, \\ \sigma(u)n = F & \text{on } \Gamma_N, \end{array} \right. \quad \begin{array}{l} (3a) \\ (3b) \\ (3c) \\ (3d) \end{array}$$

and the contact conditions

$$\left\{ \begin{array}{ll} u_n \leq 0 & \text{on } \Gamma_C, \\ \sigma_n(u) \leq 0 & \text{on } \Gamma_C, \\ \sigma_n(u) u_n = 0 & \text{on } \Gamma_C, \\ \sigma_t(u) = 0 & \text{on } \Gamma_C \end{array} \right. \quad \begin{array}{l} (4a) \\ (4b) \\ (4c) \\ (4d) \end{array}$$

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Further Topics